

Oral Diabetes Agents — Sketchy Classroom (Secretagogues, Acarbose, SGLT2)



1. GLUCAGON packets on the desks

WHAT YOU SEE

Snack packets labeled GLUCAGON scattered on the tables.

WHAT IT ENCODES

Glucagon (alpha-cell hormone) is the rescue for severe hypoglycemia when the patient can't take oral carbs — IM/SC or intranasal. It mobilizes hepatic glycogen to raise glucose within minutes.

BOARD TRAP

Glucagon fails in alcoholics or prolonged fasting (depleted glycogen stores) — give IV dextrose instead when the liver has nothing to release.

2. Cupcakes / sweets

WHAT YOU SEE

Frosted cupcakes on the back table.

WHAT IT ENCODES

Carbohydrate load = the dietary trigger. Counting carbs and the glycemic index drives prandial insulin dosing; post-prandial spikes are the target of rapid insulin, meglitinides, and acarbose.

BOARD TRAP

A simple-sugar treat is the correct treatment for conscious mild hypoglycemia (15 g fast carbs, recheck in 15 min) — not the cause to avoid in that moment.

3. Martini-glass tray (secretagogue 'bar')

WHAT YOU SEE

Woman carrying a tray of red martini glasses.

WHAT IT ENCODES

Secretagogues = sulfonylureas (glipizide, glyburide, glimepiride) and meglitinides (repaglinide, nateglinide). They close beta-cell K-ATP channels, depolarize the cell, and trigger insulin release independent of glucose.

BOARD TRAP

Because release is glucose-INDEPENDENT, sulfonylureas cause hypoglycemia even when the patient skips a meal — unlike metformin or DPP-4 inhibitors, which rarely do.

4. Woman serving (insulin release)

WHAT YOU SEE

Server handing out the drinks.

WHAT IT ENCODES

She represents forced insulin secretion. Glyburide (long-acting, renally cleared) carries the highest hypoglycemia risk, especially in the elderly and in CKD; glipizide is preferred when renal function is reduced.

BOARD TRAP

Avoid glyburide in elderly/renal patients — its active metabolites accumulate and cause prolonged hypoglycemia; choose a shorter-acting sulfonylurea or different class.

5. A-Carb table, 'maltose/disaccharide free'

WHAT YOU SEE

Boxes labeled A-Carb, 'disaccharide free'.

WHAT IT ENCODES

Acarbose/miglitol = alpha-glucosidase inhibitors. They block intestinal brush-border disaccharidases, slowing carbohydrate absorption and blunting post-prandial glucose. Taken with the first bite of each meal.

BOARD TRAP

If a patient on acarbose becomes hypoglycemic, treat with GLUCOSE (dextrose), NOT sucrose/table sugar — the blocked enzyme can't cleave sucrose to absorbable monosaccharides.

6. Empty/open A-Carb boxes (GI effect)

WHAT YOU SEE

Opened cardboard boxes spilling out.

WHAT IT ENCODES

Undigested carbs reaching the colon ferment, causing flatulence, bloating, and diarrhea — the dose-limiting GI side effects that drive poor adherence to acarbose.

BOARD TRAP

Acarbose alone does not cause hypoglycemia (no insulin secretion); hypoglycemia only appears when combined with insulin or a sulfonylurea.

7. Sweaty, pale man (hypoglycemia)

WHAT YOU SEE

Man at the desk pale, sweating, wiping his brow.

WHAT IT ENCODES

Adrenergic hypoglycemia symptoms: tremor, sweating, palpitations, anxiety (counter-regulatory epinephrine). Neuroglycopenic signs (confusion, seizure, coma) follow as glucose drops further.

BOARD TRAP

Beta-blockers MASK the adrenergic warning signs of hypoglycemia (sweating persists, tremor/tachycardia blunted) — a classic exam pitfall in diabetics on non-selective beta-blockade.

8. Passed-out hooded man on floor

WHAT YOU SEE

Man collapsed/asleep on the floor in a hoodie.

WHAT IT ENCODES

Severe neuroglycopenic hypoglycemia — loss of consciousness. Requires IV dextrose (D50) or glucagon if no IV access. Recurrent severe events cause hypoglycemia unawareness over time.

BOARD TRAP

Whipple's triad confirms true hypoglycemia: low measured glucose + consistent symptoms + relief with glucose — don't diagnose on a single low fingerstick alone.

9. Man reaching / thrown (glucose crash)

WHAT YOU SEE

Green-shirt boy lunging across the floor.

WHAT IT ENCODES

Depicts the rapid glucose crash from secretagogue overdose or missed meal. Onset is fast with rapid-acting agents (repaglinide/nateglinide), which are dosed per meal and skipped if a meal is skipped.

BOARD TRAP

Meglitinides have a SHORTER duration than sulfonylureas — lower hypoglycemia risk and a good choice for erratic eaters; 'no meal, no pill' is the rule.

10. Canadian flag (canagliflozin)

WHAT YOU SEE

Small Canadian flag on the table.

WHAT IT ENCODES

CANA-da flag cues CANAgliflozin, an SGLT2 inhibitor. SGLT2 inhibitors (-gliflozins) block proximal tubule glucose reabsorption, causing glucosuria, weight loss, BP reduction, and proven CV/renal protection.

BOARD TRAP

Canagliflozin carries an increased lower-limb amputation and fracture signal; all gliflozins risk euglycemic DKA — stop them before surgery/illness.

11. 'SALTY GLUCOSE PEPPER CORN' bag (SGLT2)

WHAT YOU SEE

Red bag reading SALTY GLUCOSE / PEPPER CORN.

WHAT IT ENCODES

Salt + glucose spilled = SGLT2 inhibition dumps sodium AND glucose in the urine. The natriuresis underlies BP lowering and heart-failure benefit (empagliflozin, dapagliflozin reduce HF hospitalization).

BOARD TRAP

SGLT2 inhibitors increase genital mycotic infections and rare necrotizing fasciitis of the perineum (Fournier gangrene), and volume depletion — not 'just a sugar pill'.

12. Scattered glucose 'corn' on table

WHAT YOU SEE

Yellow kernels spilled across the desk by the bag.

WHAT IT ENCODES

Spilled corn = urinary glucose loss from SGLT2 blockade. This explains euglycemic DKA: with glucose excreted, plasma glucose can be near-normal while ketosis develops, masking the diagnosis.

BOARD TRAP

In a gliflozin patient with malaise and high anion-gap acidosis but only mildly elevated glucose, check ketones — euglycemic DKA is easily missed because the glucose 'looks fine'.

13. RACECAR TRACK poster

WHAT YOU SEE

Wall poster of a red racecar / racetrack.

WHAT IT ENCODES

Speed cue = rapid onset/short action of meglitinides and rapid insulin analogs targeting the post-prandial 'race'. Reinforces matching fast agents to meal-time glucose excursions.

BOARD TRAP

Don't pair a rapid prandial agent with a missed meal — the 'fast car with no fuel' is the setup for post-dose hypoglycemia.

14. Maracas / shaking child (tremor)

WHAT YOU SEE

Child shaking maracas in the lower-left.

WHAT IT ENCODES

Shaking = the tremor of catecholamine-driven hypoglycemia and a memory hook for adrenergic symptoms. Counter-regulation (glucagon then epinephrine, cortisol, GH) defends against falling glucose.

BOARD TRAP

In long-standing type 1 with autonomic neuropathy, the glucagon AND epinephrine responses fail — leading to dangerous hypoglycemia unawareness.
